

Load Interconnection System Impact Study Report

For Queue Project Q12-0164

Mayo 69 kV
August 20, 2013

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q12-0164. The Interconnection Customer proposes a 406.5 MW large load (data center) facility interconnecting at the Mayo 69 kV substation in the North Carolina area. The request was received on October 10, 2012. The study evaluated system intact and single contingency conditions using the regional planning case, including 133 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 20 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$160,606,000.

3. Network Impacts

3.1 Load Deliverability Analysis

1. Facility: Honeyville 138 kV/69 kV Autotransformer #1
Contingency: Malaspina – Ols Energy Chino 138kV Line
Loading: 199.0% (Rating: 88 MVA)
DFAX: 27.7%
Pre-project loading: 70.7%
2. Facility: Honeyville 138/69 kV XFMR Unit 2
Contingency: Ols Energy Chino – Malaspina 138kV L
Loading: 199.0% (Rating: 88 MVA)
DFAX: 27.7%
Pre-project loading: 70.7%
3. Facility: Honeyville 138/69 kV Transformer Bank Unit 3
Contingency: Ols Energy Chino – Malaspina Substation 138kV
Loading: 199.0% (Rating: 88 MVA)
DFAX: 27.7%
Pre-project loading: 70.7%
4. Facility: Honeyville – Brimley 138kV line
Contingency: Mecklenburg Power Station – Kerr Dam 345kV
Loading: 172.2% (Rating: 219 MVA)
DFAX: 25.4%
Pre-project loading: 124.8%
5. Facility: Ols Energy Chino – Malaspina 138kV Line
Contingency: Mecklenburg Power Sta – Person 345kV
Loading: 166.9% (Rating: 200 MVA)
DFAX: 42.6%
Pre-project loading: 80.3%
6. Facility: Pine River Alte – Ols Energy Chino 138kV
Contingency: Mecklenburg Power Sta – Person 345kV line
Loading: 147.1% (Rating: 222 MVA)

DFAX: 35.7%

Pre-project loading: 81.9%

7. Facility: Malaspina 138-69 kV Auto Bank 1

Contingency: Malaspina 138/69kV Transformer #2

Loading: 144.8% (Rating: 75 MVA)

DFAX: 11.3%

Pre-project loading: 84.1%

8. Facility: Malaspina 138-69 kV Auto

Contingency: Malaspina 138/69 kV Auto Unit 1

Loading: 144.8% (Rating: 75 MVA)

DFAX: 11.3%

Pre-project loading: 84.1%

9. Facility: State School – Pine Riv Alte 138kV

Contingency: Person 138 kV/1 kV Transformer Bank Bank 1

Loading: 135.9% (Rating: 244 MVA)

DFAX: 24.5%

Pre-project loading: 95.0%

10. Facility: Brimley – Mecklenburg Power Station 138kV

Contingency: Mecklenburg Power Sta – Kerr Dam 345kV

Loading: 133.6% (Rating: 221 MVA)

DFAX: 22.8%

Pre-project loading: 91.7%

11. Facility: Honeyville – Tecumseh Energy Ctr 138kV L

Contingency: Brimley – Honeyville 138kV

Loading: 125.7% (Rating: 206 MVA)

DFAX: 40.5%

Pre-project loading: 45.6%

12. Facility: Grady 138/69 kV Auto

Contingency: Grady 138/69kV TX

Loading: 115.6% (Rating: 80 MVA)

DFAX: 13.6%

Pre-project loading: 46.8%

13. Facility: Grady 138 kV/69 kV TX

Contingency: Grady 138/69 kV Transformer Bank

Loading: 115.6% (Rating: 80 MVA)

DFAX: 13.6%

Pre-project loading: 46.8%

14. Facility: Grady 138/69 kV XFMR

Contingency: Grady 138-69 kV Auto T1

Loading: 115.6% (Rating: 80 MVA)

DFAX: 13.6%

Pre-project loading: 46.8%

15. Facility: Urban – American 345kV

Contingency: West Waynesboro – Chambersburg Substation 345kV

Loading: 109.8% (Rating: 1,270 MVA)

DFAX: 7.7%

Pre-project loading: 107.4%

16. Facility: Person – Pine Riv Alte 138kV

Contingency: Mecklenburg Power Sta – Person 345kV

Loading: 107.8% (Rating: 412 MVA)

DFAX: 12.4%

Pre-project loading: 95.6%

17. Facility: Front Royal Substation – Urban 345kV

Contingency: Urban – International Paper Valliant Ok Substation 345kV Line

Loading: 106.6% (Rating: 1,036 MVA)

DFAX: 6.2%

Pre-project loading: 104.1%

18. Facility: Alligator 500-345 kV TX

Contingency: Enka – Thiele Kaolin Reedy Crk 345kV

Loading: 105.9% (Rating: 446 MVA)

DFAX: 5.7%

Pre-project loading: 100.8%

19. Facility: State School 345-138 kV XFMR

Contingency: Mecklenburg Power Sta – Person 345kV

Loading: 102.6% (Rating: 596 MVA)

DFAX: 19.1%

Pre-project loading: 89.6%

20. Facility: Ridgeley – Spindle 345kV L

Contingency: E Devon – Thiele Kaolin Reedy Crk 345kV

Loading: 101.0% (Rating: 1,287 MVA)

DFAX: 6.5%

Pre-project loading: 98.9%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Honeyville 138 kV/69 kV Autotransformer #1	Replace transformer with higher-rated unit	\$7,462,000	100% (\$7,462,000)
Honeyville 138/69 kV Autotransformer	Replace transformer with higher-rated unit	\$4,936,000	100% (\$4,936,000)
Honeyville 138/69kV Auto	Replace transformer with higher-rated unit	\$6,001,000	100% (\$6,001,000)
Honeyville – Brimley 138kV line	Rebuild line with higher-capacity conductor	\$33,941,000	100% (\$33,941,000)
Ols Energy Chino – Malaspina 138kV Line	Rebuild line with higher-capacity conductor	\$8,954,000	100% (\$8,954,000)
Pine River Alte – Ols Energy Chino 138kV	Rebuild line with higher-capacity conductor	\$22,095,000	100% (\$22,095,000)
Malaspina 138-69 kV Auto Bank 1	Replace transformer with higher-rated unit	\$4,009,000	100% (\$4,009,000)
Malaspina 138-69 kV Auto	Replace transformer with higher-rated unit	\$2,825,000	100% (\$2,825,000)
State School – Pine Riv Alte 138kV	Rebuild line with higher-capacity conductor	\$3,988,000	100% (\$3,988,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Brimley – Mecklenburg Power Station 138kV Line	Rebuild line with higher-capacity conductor	\$3,713,000	100% (\$3,713,000)
Honeyville – Tecumseh Energy Ctr 138kV line	Rebuild line with higher-capacity conductor	\$8,204,000	100% (\$8,204,000)
Grady 138/69 kV Auto	Replace transformer with higher-rated unit	\$7,543,000	100% (\$7,543,000)
Grady 138 kV/69 kV TX	Replace transformer with higher-rated unit	\$11,083,000	100% (\$11,083,000)
Grady 138/69 kV XFMR	Replace transformer with higher-rated unit	\$5,930,000	100% (\$5,930,000)
American – Urban 345kV	Reconductor line	\$26,187,000	11% (\$2,890,000)
Person – Pine Riv Alte 138kV	Reconductor line	\$3,705,000	100% (\$3,705,000)
Front Royal Substation – Urban 345kV	Reconductor line	\$25,337,000	16% (\$4,095,000)
Alligator 500 kV/345 kV TX	Replace transformer with higher-rated unit	\$17,501,000	9% (\$1,551,000)
State School 345/138kV XFMR	Replace transformer with higher-rated unit	\$15,281,000	97% (\$14,761,000)
Ridgeley – Spindle 345kV L	Reconductor line	\$18,274,000	16% (\$2,920,000)
	Total allocated to project		\$160,606,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.